Computer Vision Peer Review

Max Moir
University of Canterbury
Christchurch, New Zealand
maxwmoir@gmail.com

May 19, 2026

1 Summary

This paper states a proposal for the development of a general purpose robot for assisting in the caretaking of elderly and disabled people. The goal of the paper is to develop a robot that can be adapted to tasks such as navigation, object recognition and avoidance, pose recognition and communication. It is stated that the main focus of the project is the transport of hospital related items.

The paper then provides background and preliminary examples related to the area, citing two projects similar to the one proposed.

It outlines the technical approach that may be taken to build an example robot to meet its criteria. The proposed implementation suggests using Microsoft 360 Connect camera to detect and follow ArUco markers. It also implies it will use both a Raspberry Pi3 and an Arduino Mega 2560 Microcontroller.

It then states how said robot may be evaluated and tested.

The author also attempts to address ethical concerns and potential future research direction.

It cites two papers that detail applications similar to the intended goal. While the chosen application area is well motivated, the paper does not have sufficient technical depth, experimental results or testing validation.

2 Decision

The final decision of the review is to reject this proposal. The paper requires major revision before it will be in a publishable state, due to the concerns outlined in the following sections. The paper has unclear contributions, insufficient novelty, and a lack of any completed experiments. Therefore it does not meet the standards for publication of a computer vision research paper. Despite the execution falling short in multiple regards, the topic is well motivated and could make for an interesting paper if done correctly.

3 Major Concerns

A first major concern with this paper is that it has unclear contributions. It frequently mixes implemented functionality with speculative goals, making it difficult to understand what has been achieved. For exampl This is not consistent with the goal of the paper to build a useable outcome, and it is unclear what the research contributions of this paper are. It follows naturally from this that the evaluation has a lack of experimental or computational data. The entire evaluation and testing section is purely hypothetical, and resemble a brainstorm more than serious scientific discussion. and presents no contributions, let alone novel ones. In order to resolve this concern, the author would need to undergo extensive further work, including developing the proposed robot and conducting new experiments. This is a major factor in the decision to reject this paper. Further work should include quantitative evaluation of the developed outcome such as clearly defined success metrics, extensive testing and benchmarks (e.g. for navigational performance or tracking accuracy).

Additionally, the design plan detailed in the paper has a lack technical depth. Even if the outcome were created according to the specification, it would likely be inadequate in terms of novelty. It lists a rough list of potential hardware that might be used to create the robot, but does not go into detail about the system design decisions made for the project. To resolve this concern, the approach section would need significant additions. For example, it should include discussion of the computer vision pipeline, navigational methedology and

Another concern is the minimal citation of previous work. Although there are a number of referenced papers in the bibliography, the author only addresses a couple of these. The references that are addressed are not evaluated in significant depth,

Finally, the paper has an underdeveloped ethical and safety implications section. It fails to address potential safety concerns and brushes off ethical issues as "beyond the scope of this paper". It mentions potential privacy concerns, but does not detail how these will be addressed. Physical safety concerns have also not been addressed, such as collision detection or reliability of the system. This is especially important for technology in the healthcare sector, and should be discussed in a publishable paper.

4 Minor Concerns

In addition to the major concerns about this paper stated above, there are several minor concerns that require revisions. First of all, the paper has many spelling and grammatical mistakes. For this paper to reach a publishable state, these mistakes, as well as any others within the text, must be resolved. A non-comprehensive list of grammatical inaccuracies is as follows:

1. Spelling Mistakes
   - *"(partivuarly repetitive and simple ones)"* → particularly
   - *hospital personell* → personnel
2. Missing words and articles
   - *"Arduino mega 2560 is used for the large number of digital pins"*
   - *items an elderly has*
3. Incorrect plurals
   - *"items an elderly has"*
4. Missing punctuation
   - *"No obstacle detection is performed, hence the robot will bump into any objects The robot will stop if it cannot find the marker."*
5. Corrupted text
   - *"used.*
     –¿ arco principes of opereation"

In addition to the grammatical errors, the paper is generally poorly worded. It contains many rambling or incomplete sentances and awkward phrasings. For example, *"Due to the nature of robots, however, repetitive tasks without the need for complex decision making are desirable as opposed to ones that require a significant level of decision making (medical diagnosis for instance)."* The author also uses redundant definitions such as: *"The robot must be seamless and operate effectively within it's environment even to the extent of operating effectively within an unfamiliar environment that it was not designed for in a way that does not compromise human safety"*. This sentance could be significantly shorted. Thourough proofreading would resolve this concern.

The reference usage throughout the paper is also often incorrect or inconsistent. Some examples of this include:

- Empty citation placeholder: *"[] illustrates a robot capable..."*
- Incorrect citation spacing: *"coming 5 years.[2]"*
- Footnote formatting issue: *"With the rise in technology [1]"*

- Unreferenced claims regarding healthcare staffing and demographics

The paper also has multiple inconsistencies in relation to formatting. The paragraph length and spacing between paragraphs is in many points inconsistent. The Subsections are formatted strangely, for example, the Approach section starts with a hardware list before entering a very short Overall System subsection. The organization of sections is also Figure 1 is missing entirely despite its present label. Overall significant proofreading and restructuring of the paper would be necessary for it to meet a publishable standard.

5 Conclusion